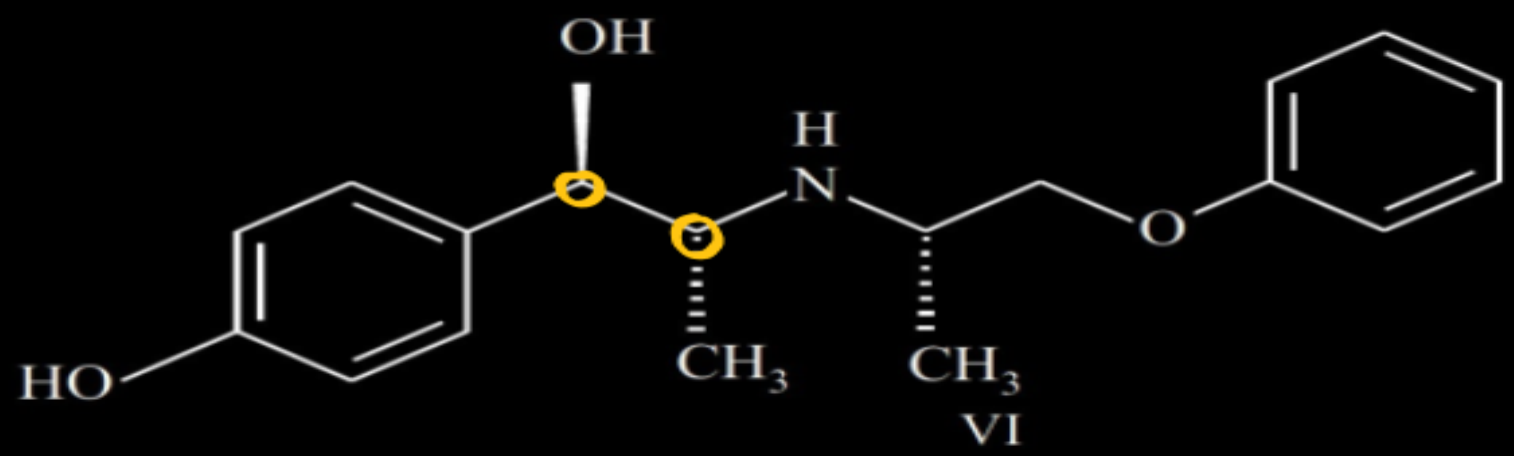
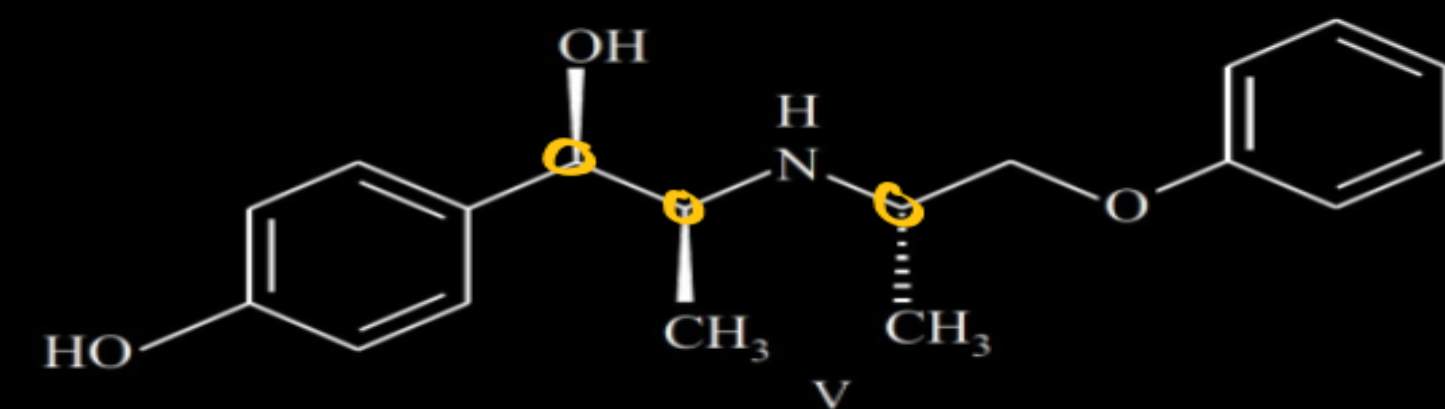
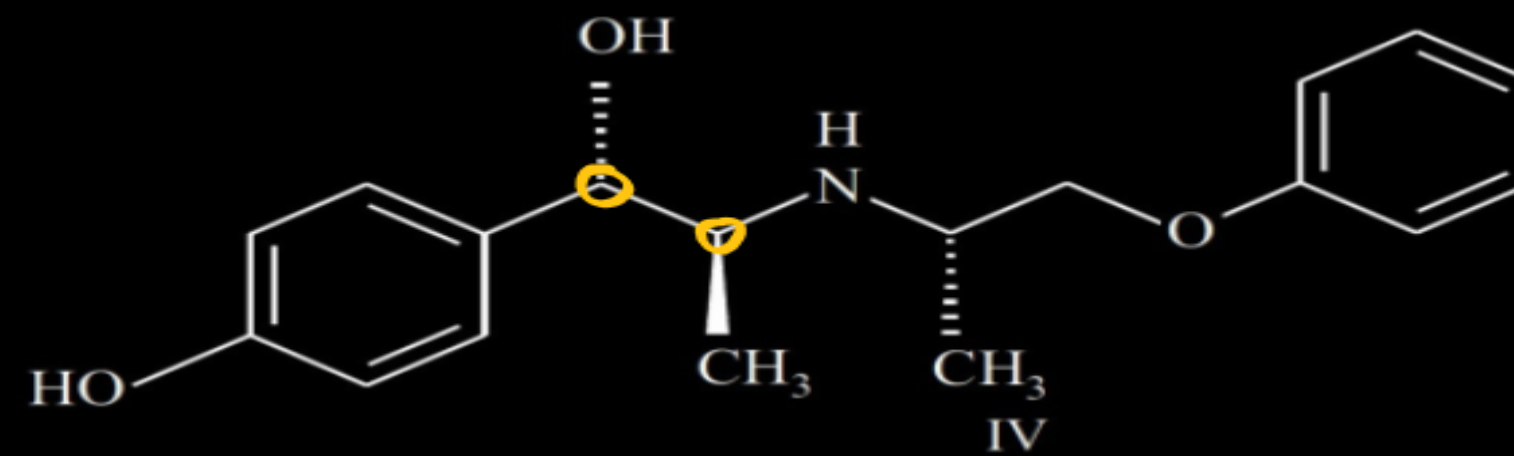
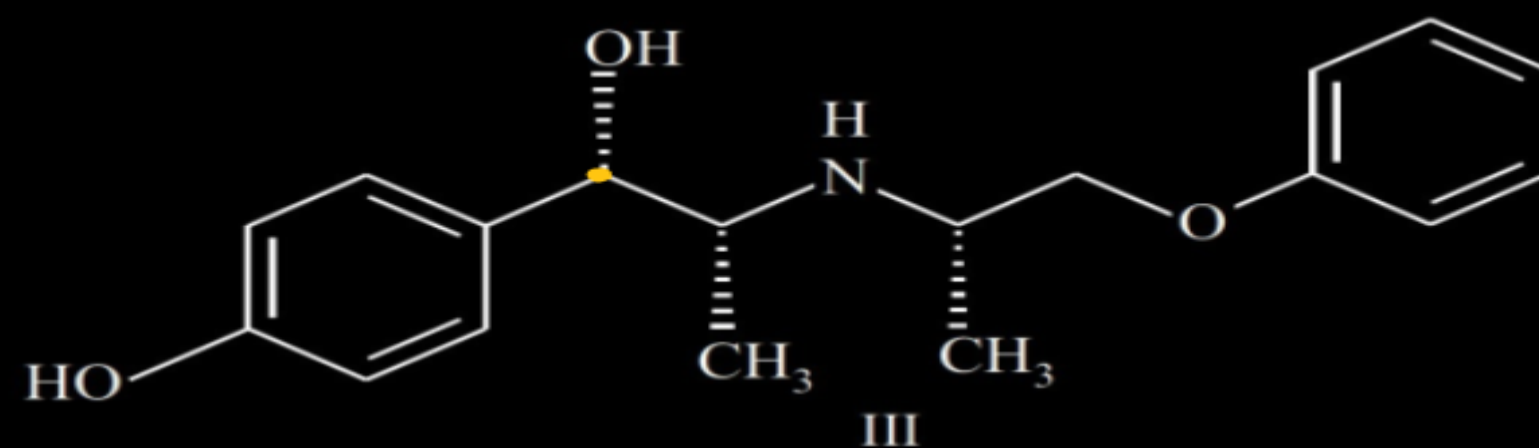
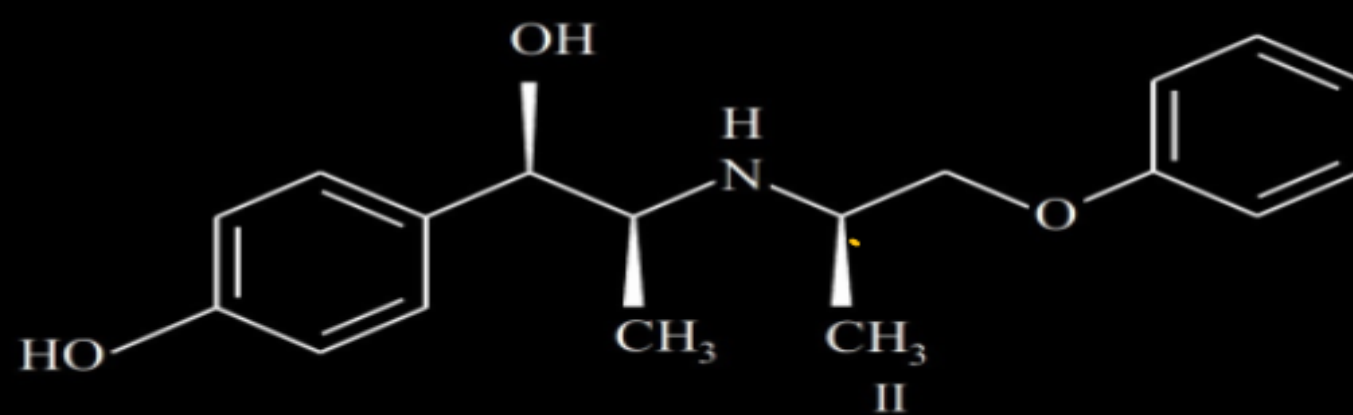
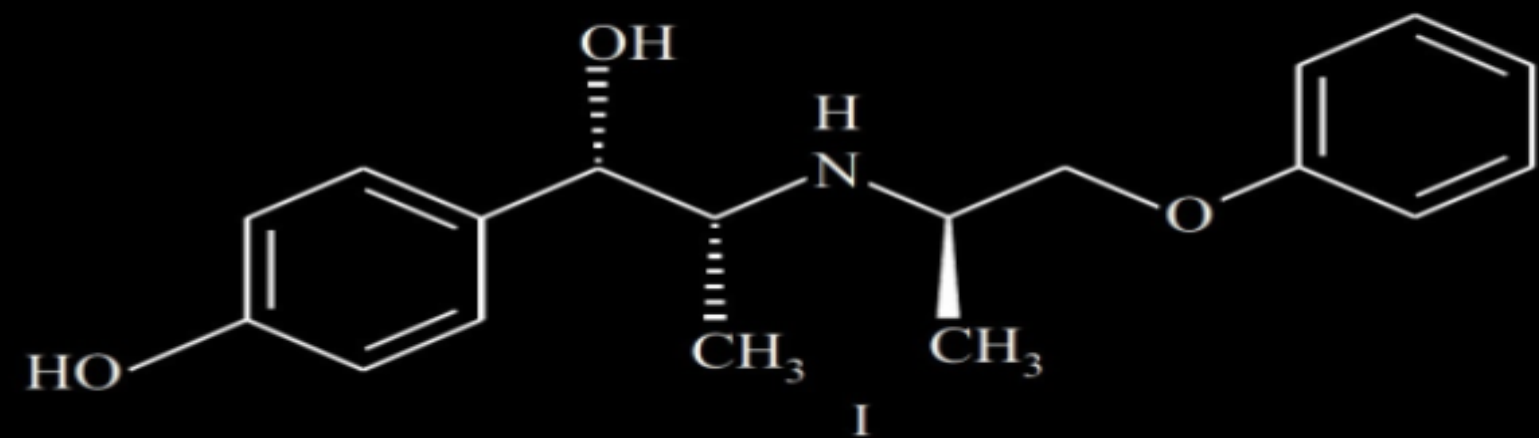


CATS-PLUS

2022

Consider the following six structures:



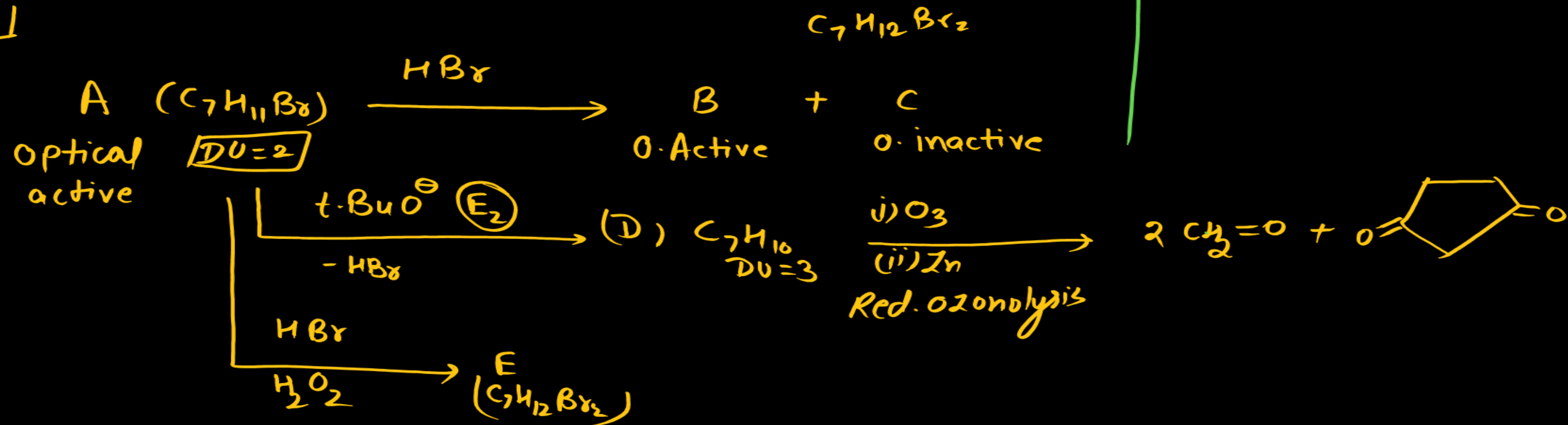
Establish the stereochemical relationship between:

- (a) I and II, (b) III and IV, (c) II and III, (d) I and V, (e) IV and VI.

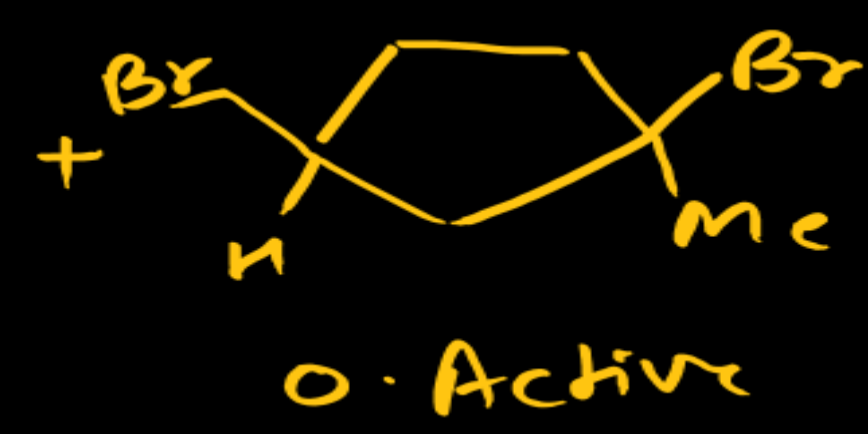
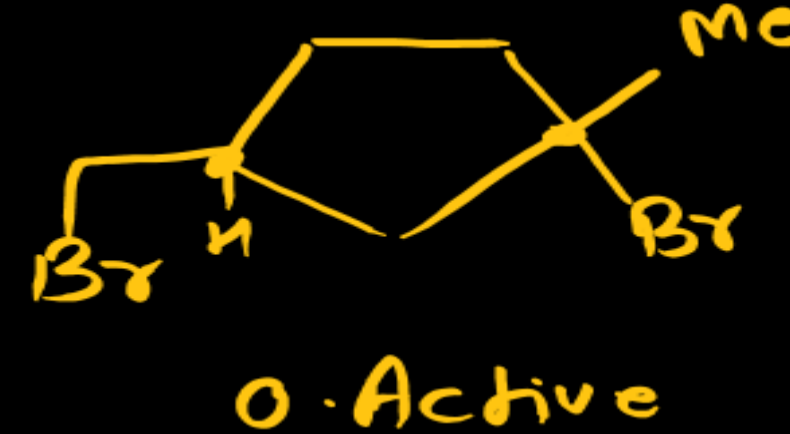
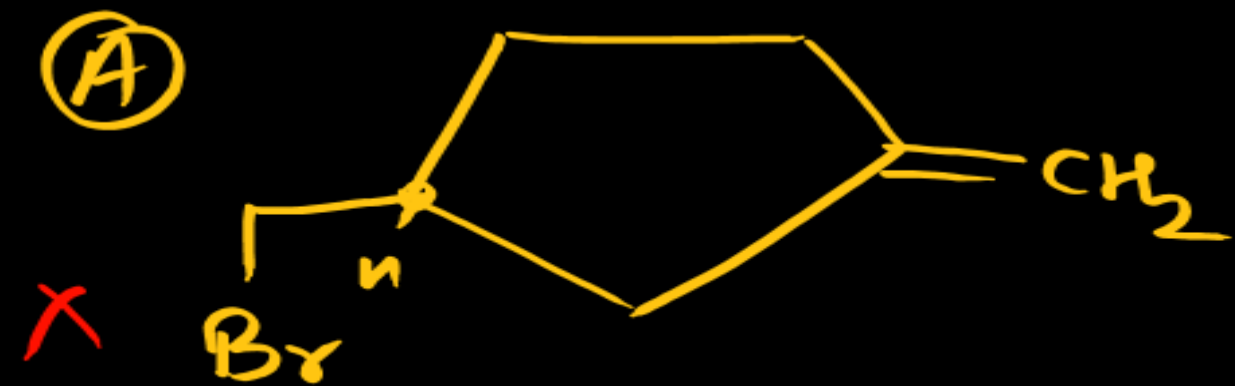
- (a) Diastereomers
 (b) Diastereomers.
 (c) Enantiomers.
 (d) Enantiomers.
 (e) Diastereomers

Q. An optically active compound A has molecular formula $C_7H_{11}Br$. A reacts with HBr, in absence of peroxide to yield isomeric products, B and C with molecular formula $C_7H_{12}Br_2$. Compound B is optically active; C is not. Treating A with one mole of potassium-t-butoxide yields D (C_7H_{10}). Subjecting one mole of D to ozonolysis followed by treatment with Zn and water yield two moles of formaldehyde and one mole of 1,3-cyclopentandione. A on addition of HBr in presence of H_2O_2 produces another isomer of B which is E, and shows diastereomerism. Deduce structure of A to E.

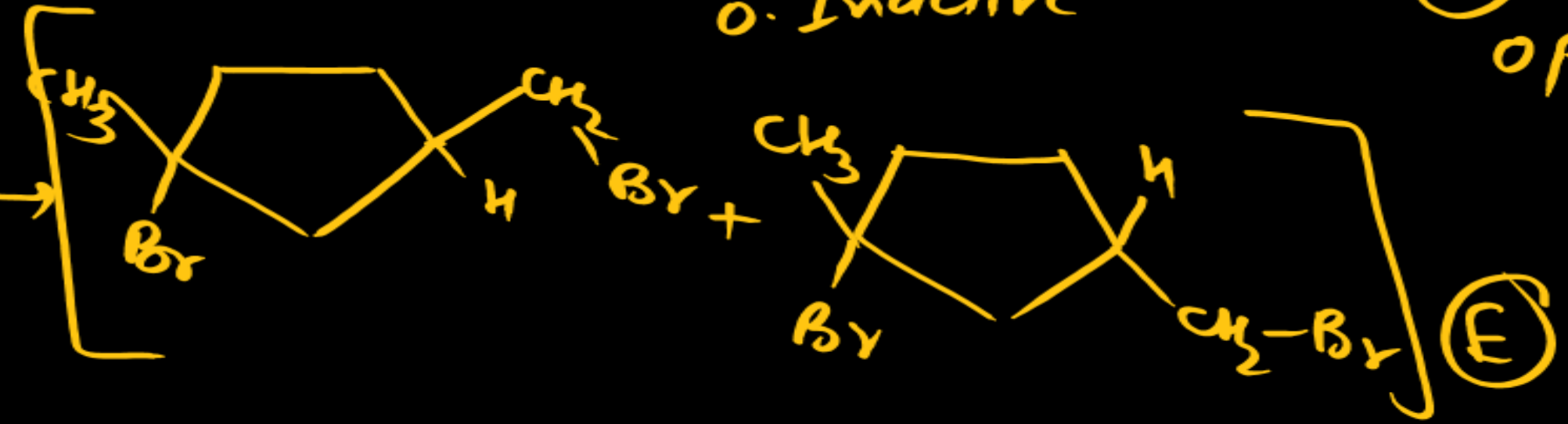
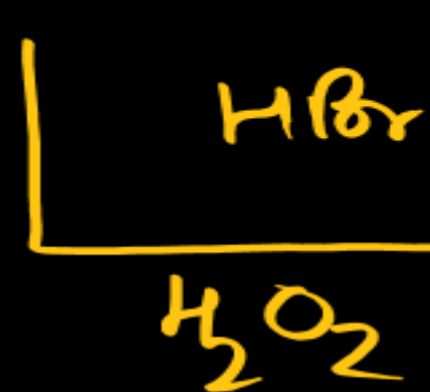
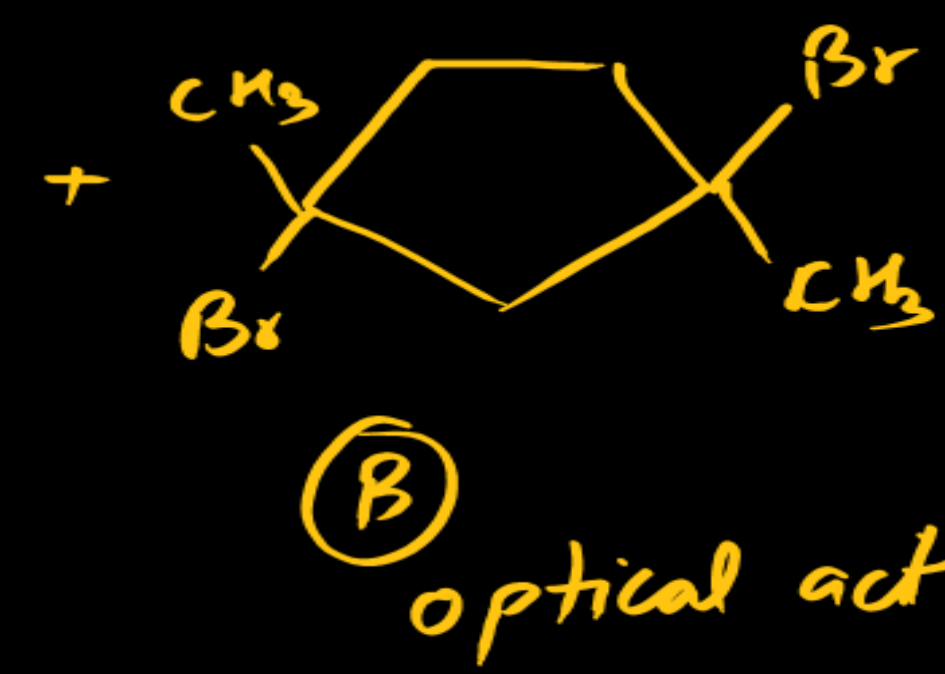
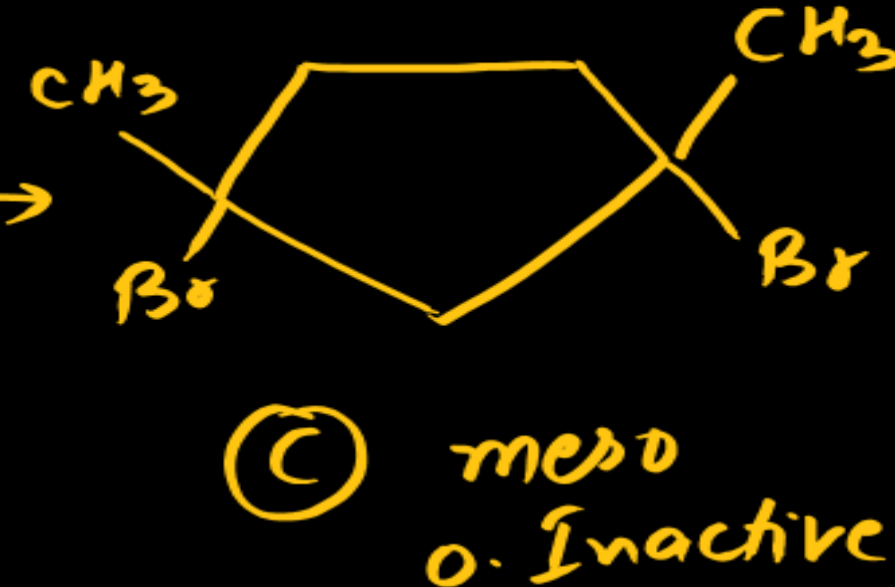
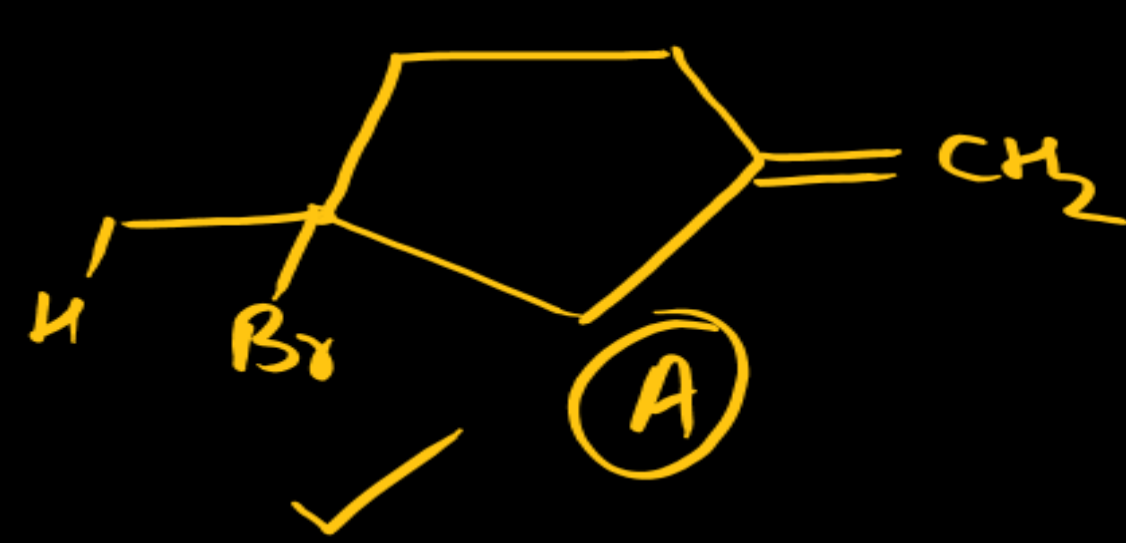
Sol



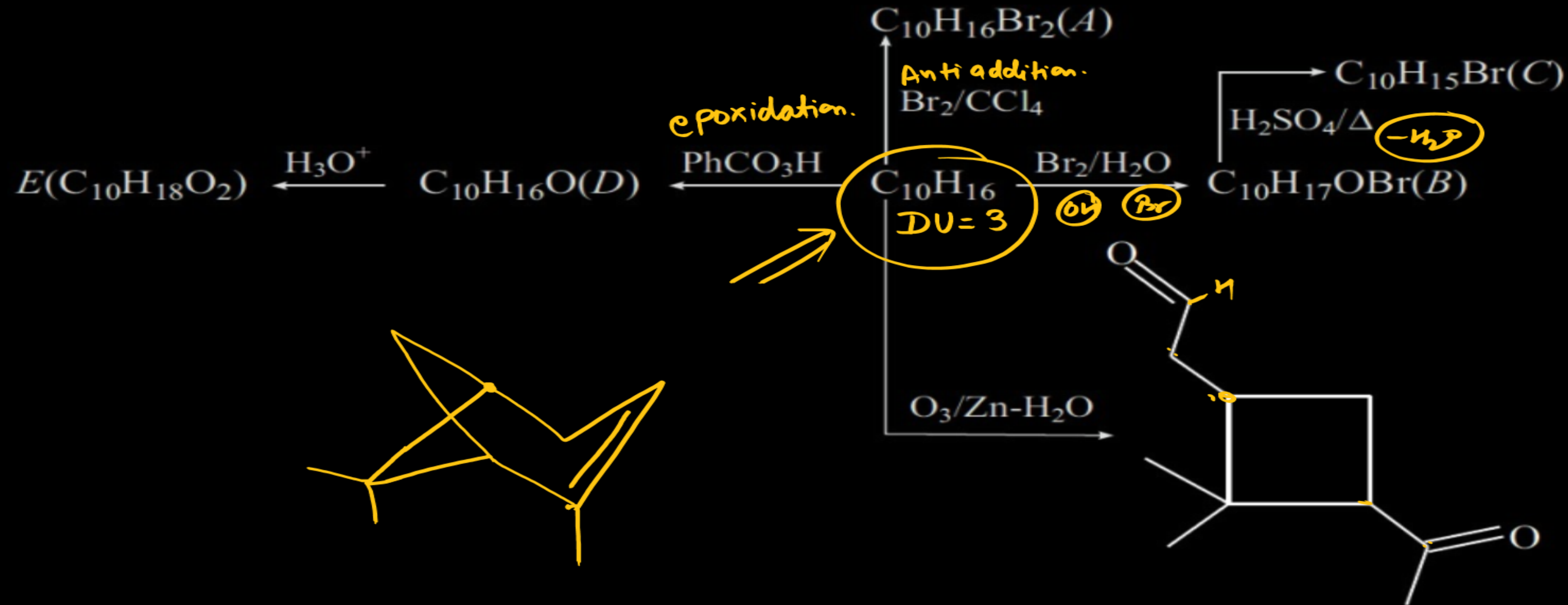
$t\text{-BuO}^-$

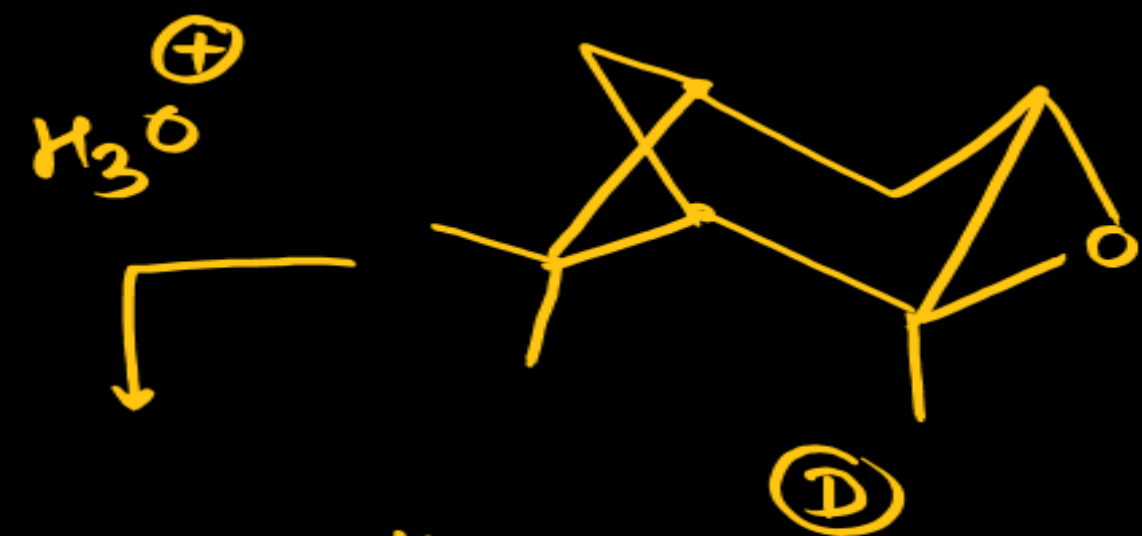
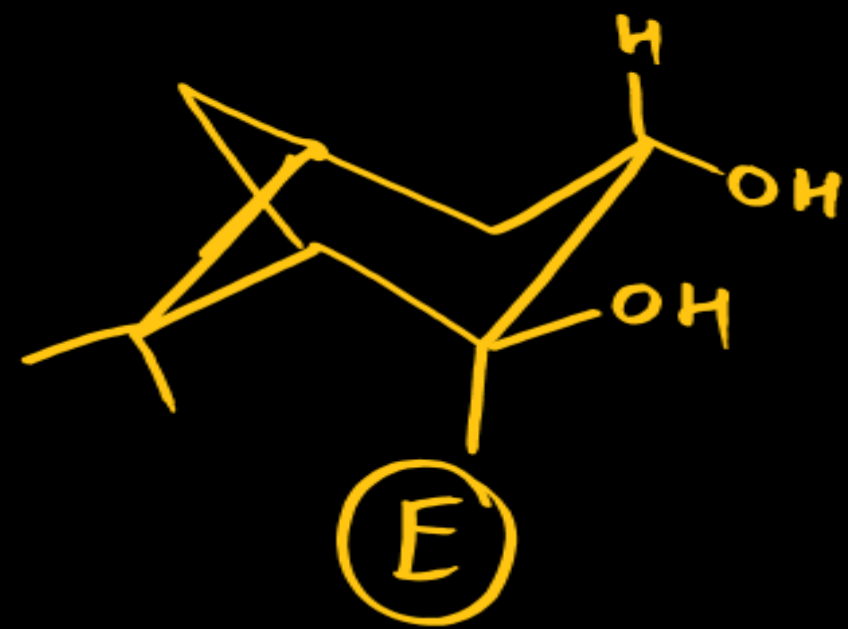


or



Identify A to E (Provide structures)





$PhCO_3H$



Anti
add..

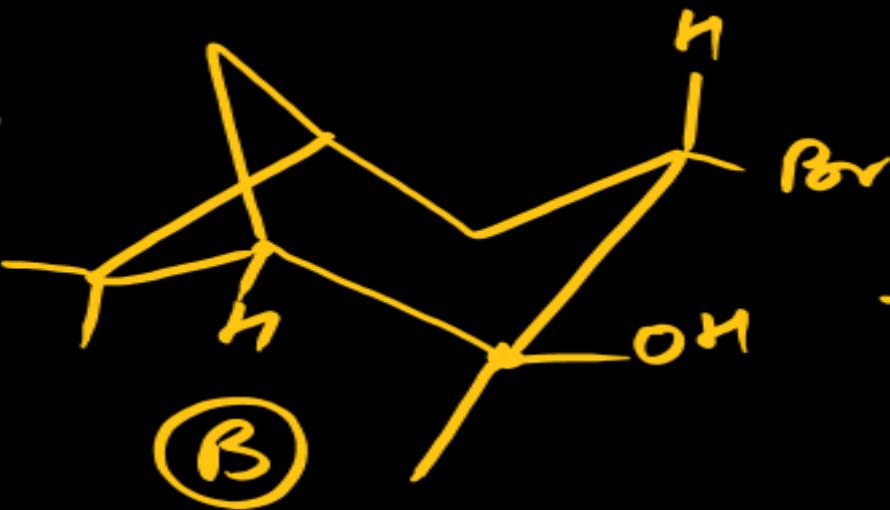
Br_2



$Br_2 + H_2O$

$HOBr$

Anti
addition.

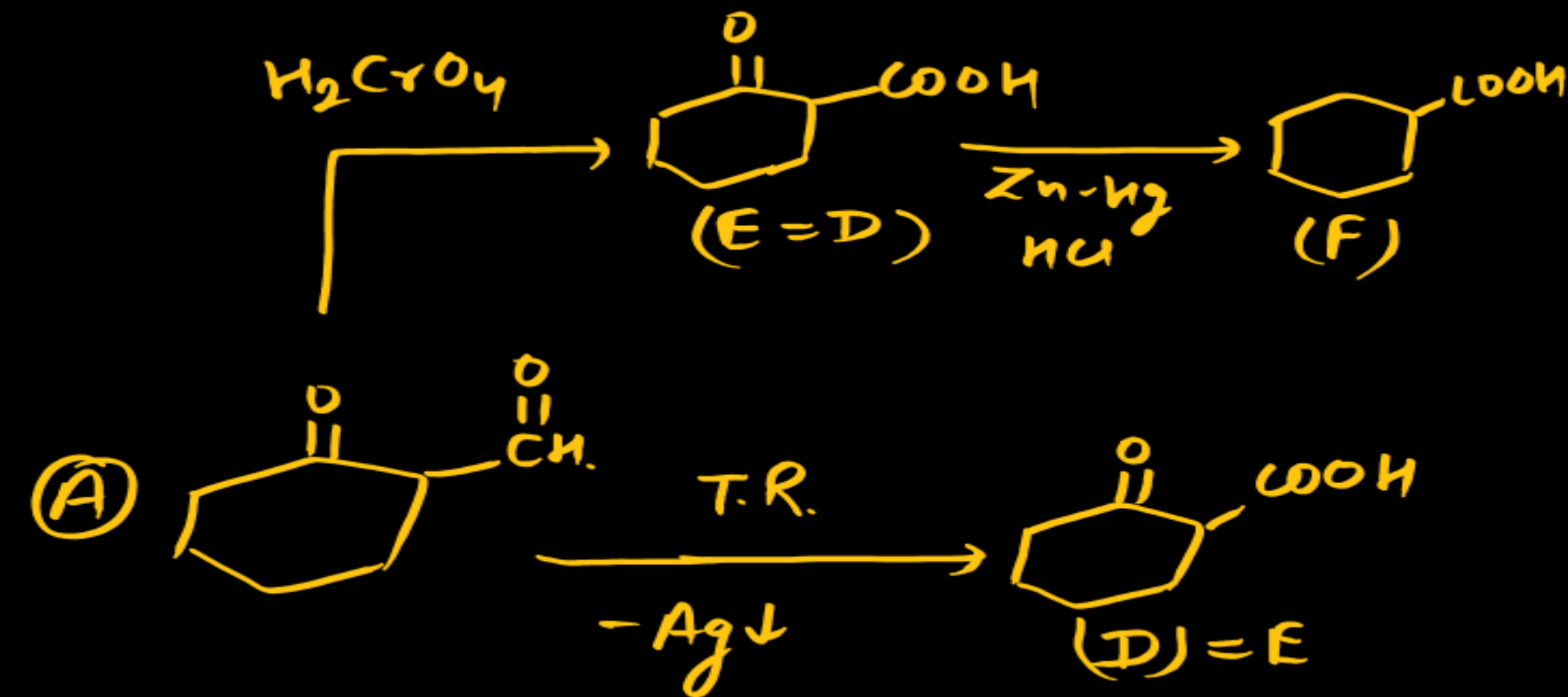
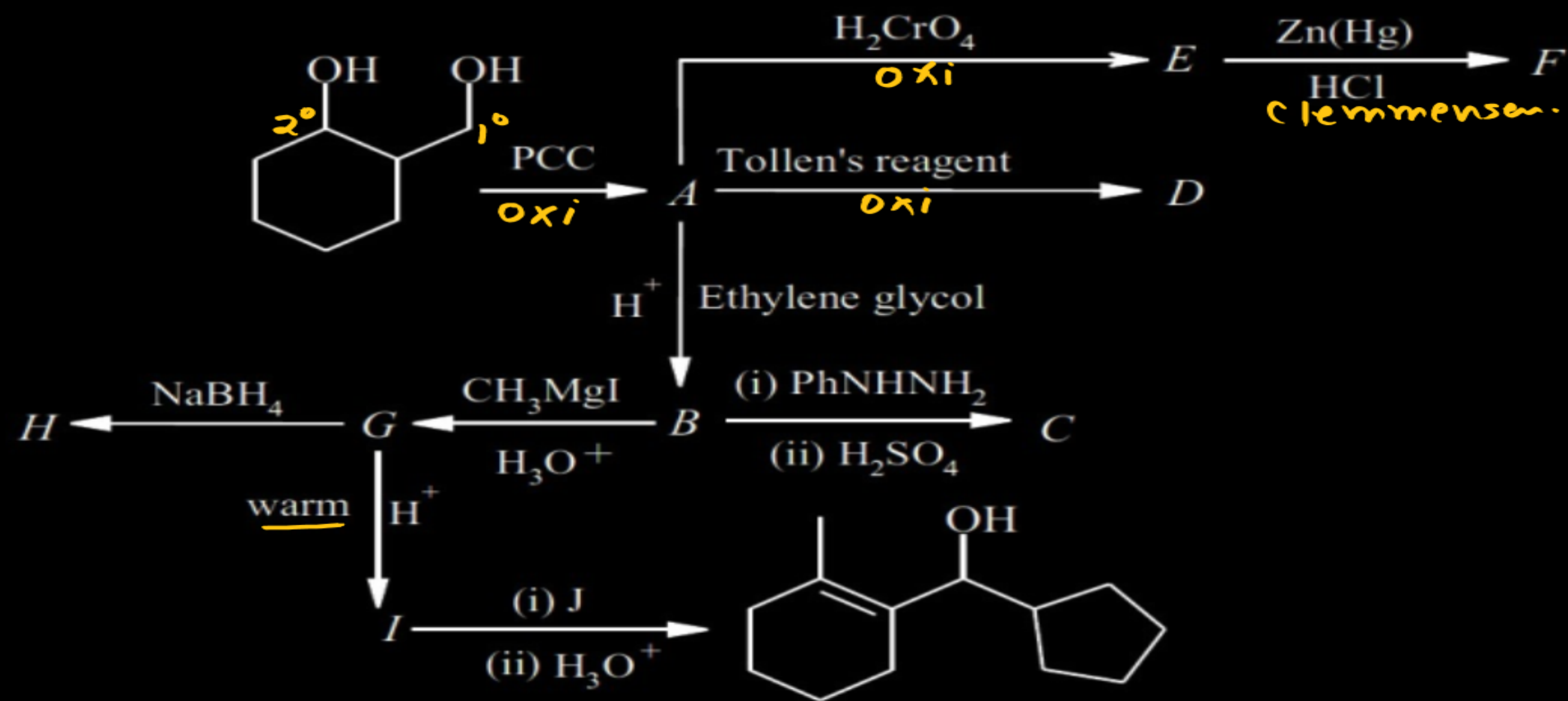


$H_2SO_4 \Delta$

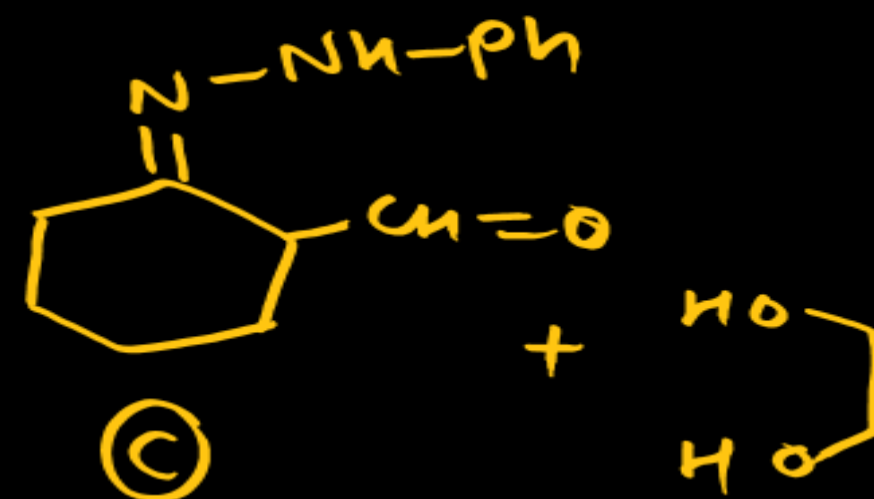
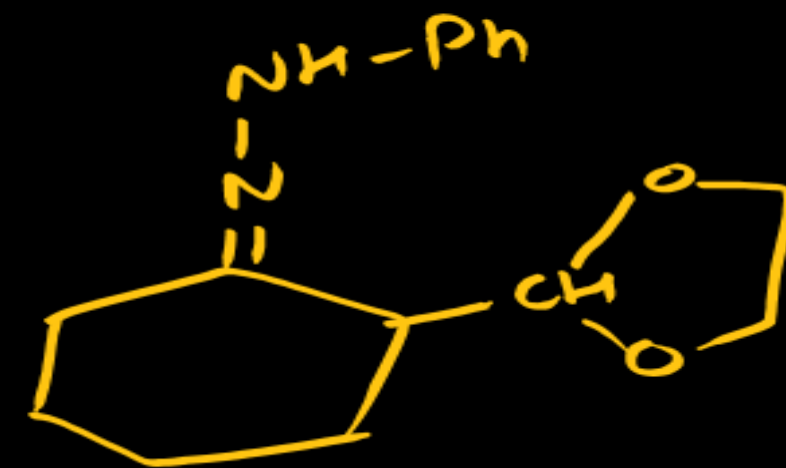
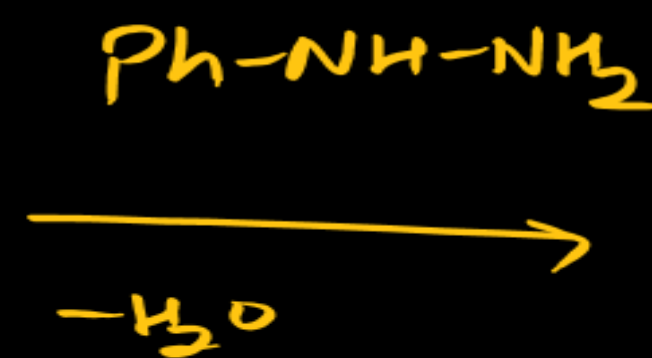
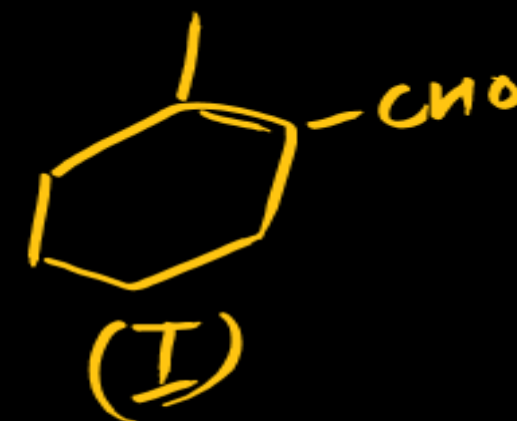
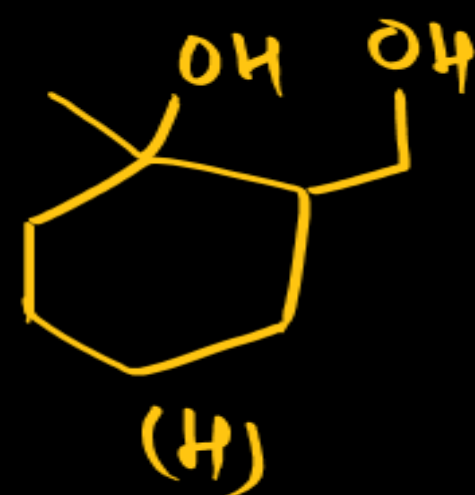
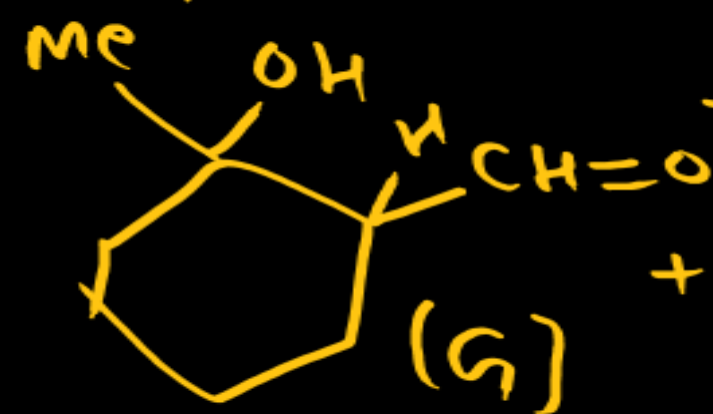
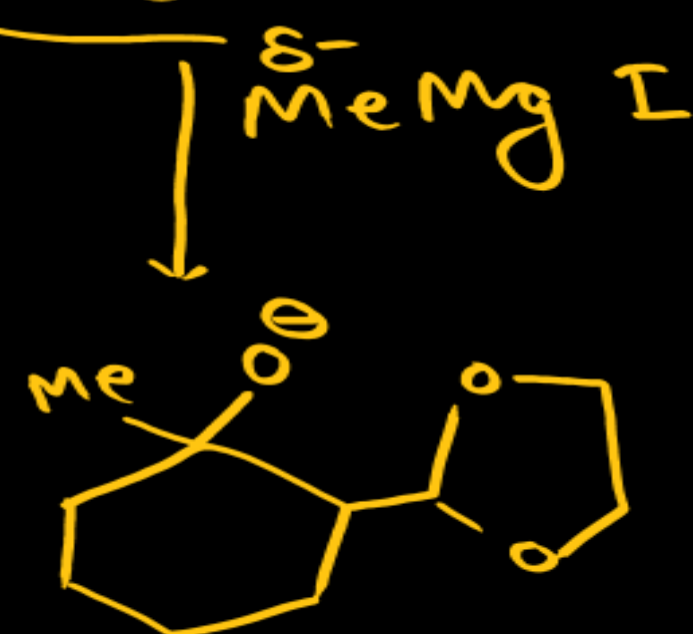
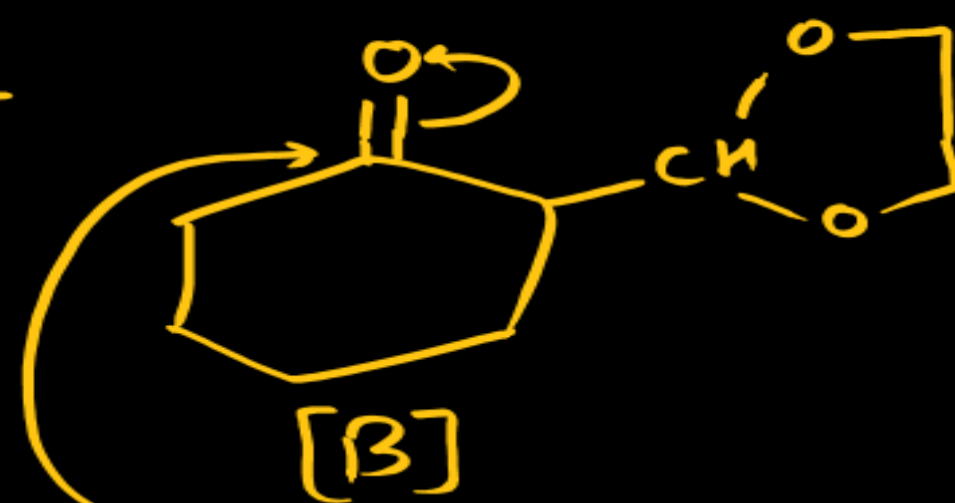
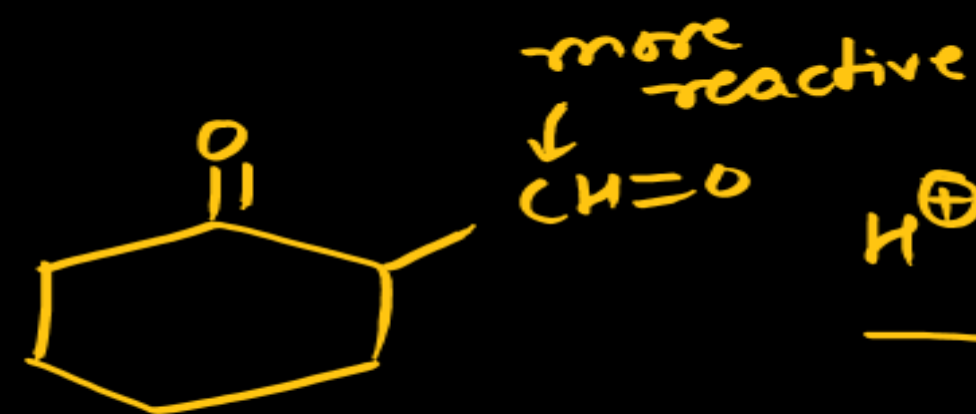
$-H_2O$

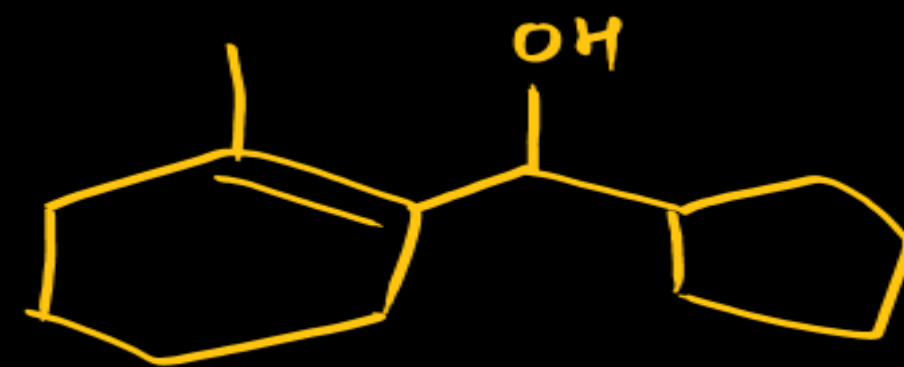
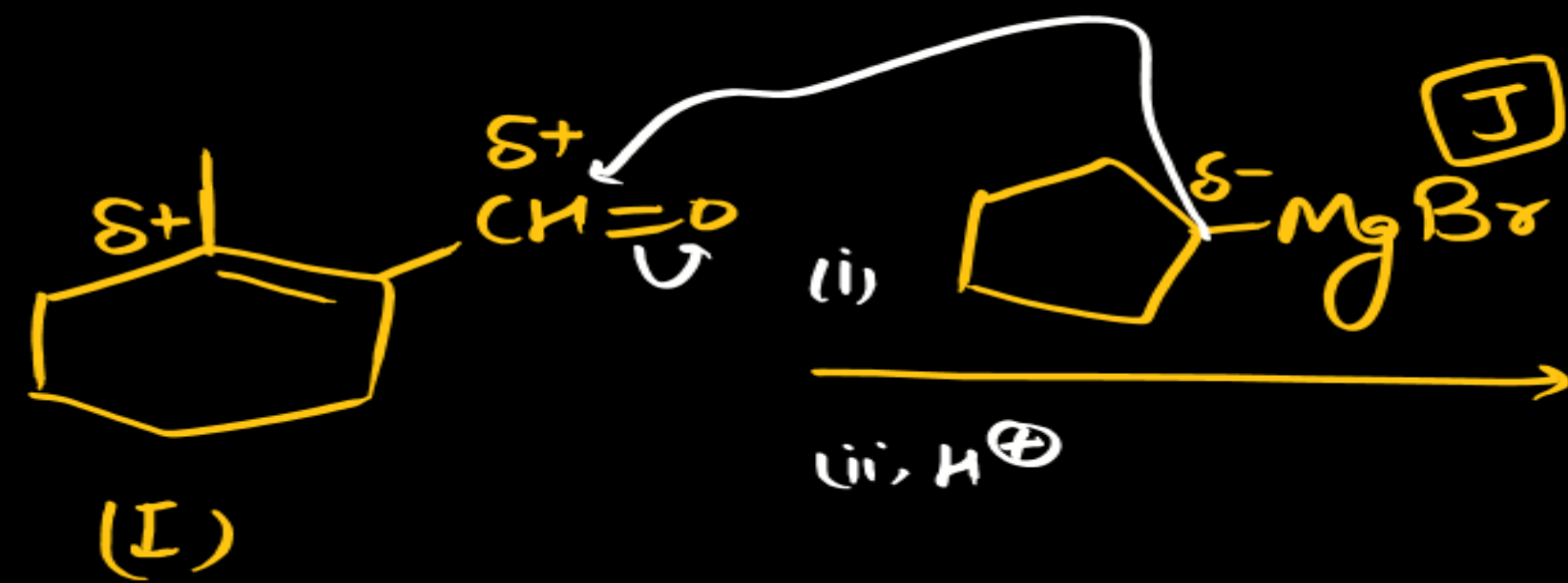


The following road-map problem centers on the structure and properties of A, a key intermediate in these reaction. Give structures for the compounds A through J:

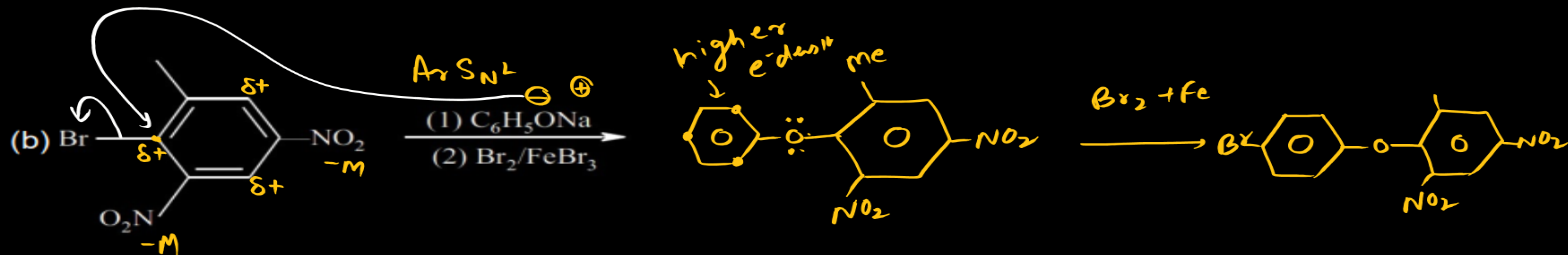


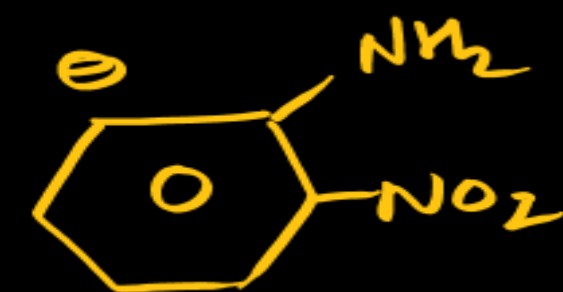
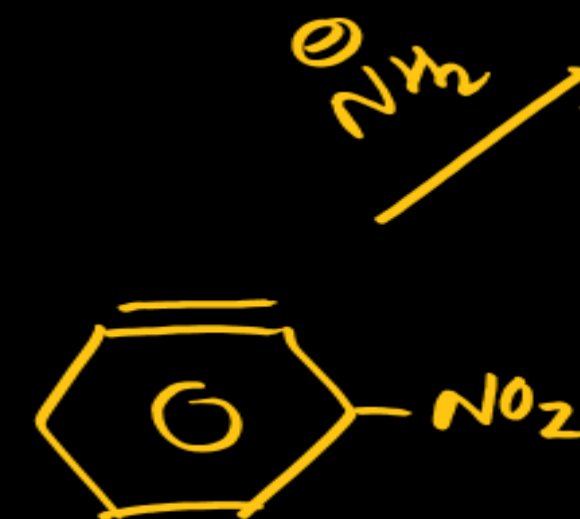
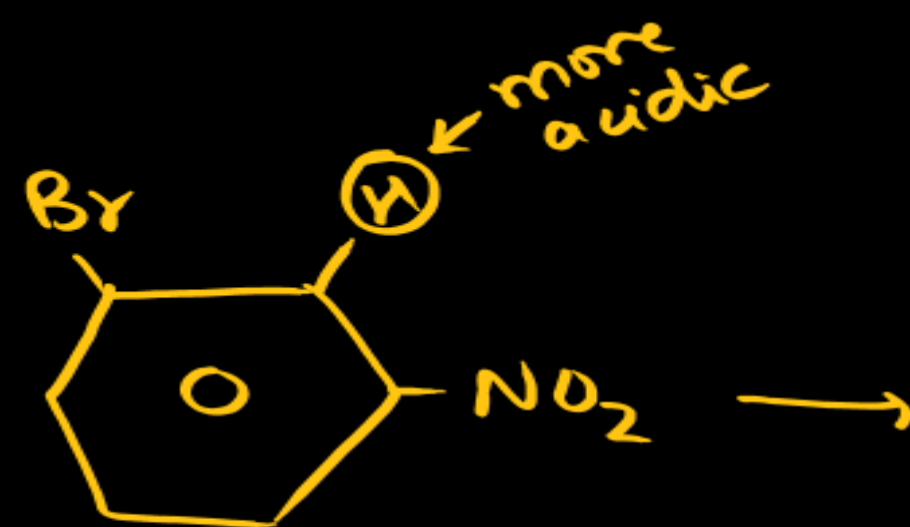
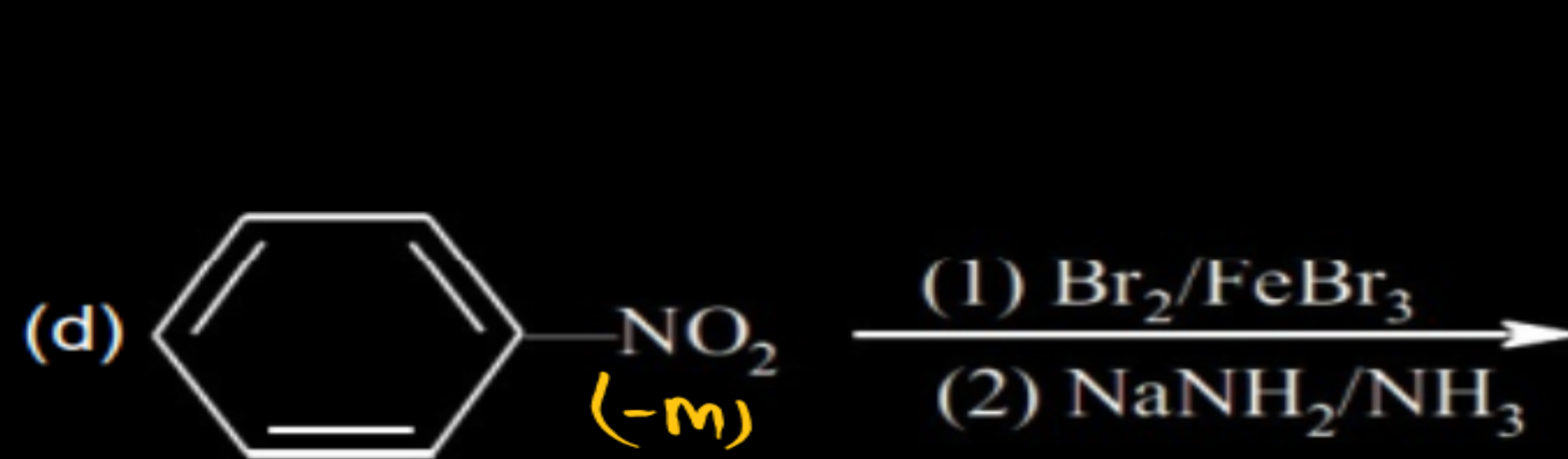
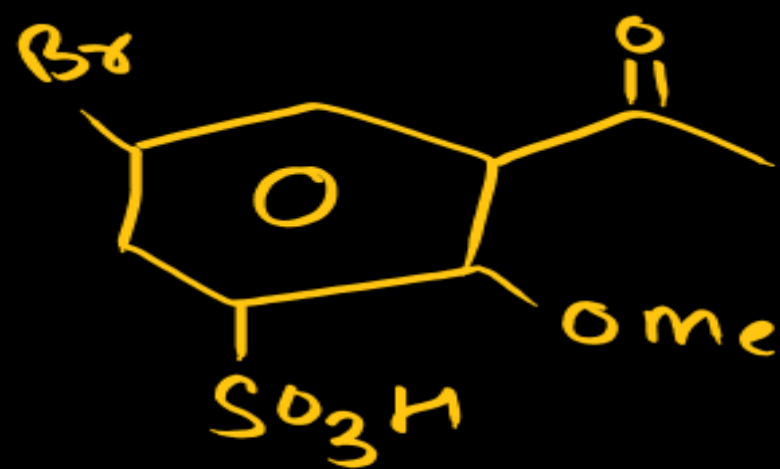
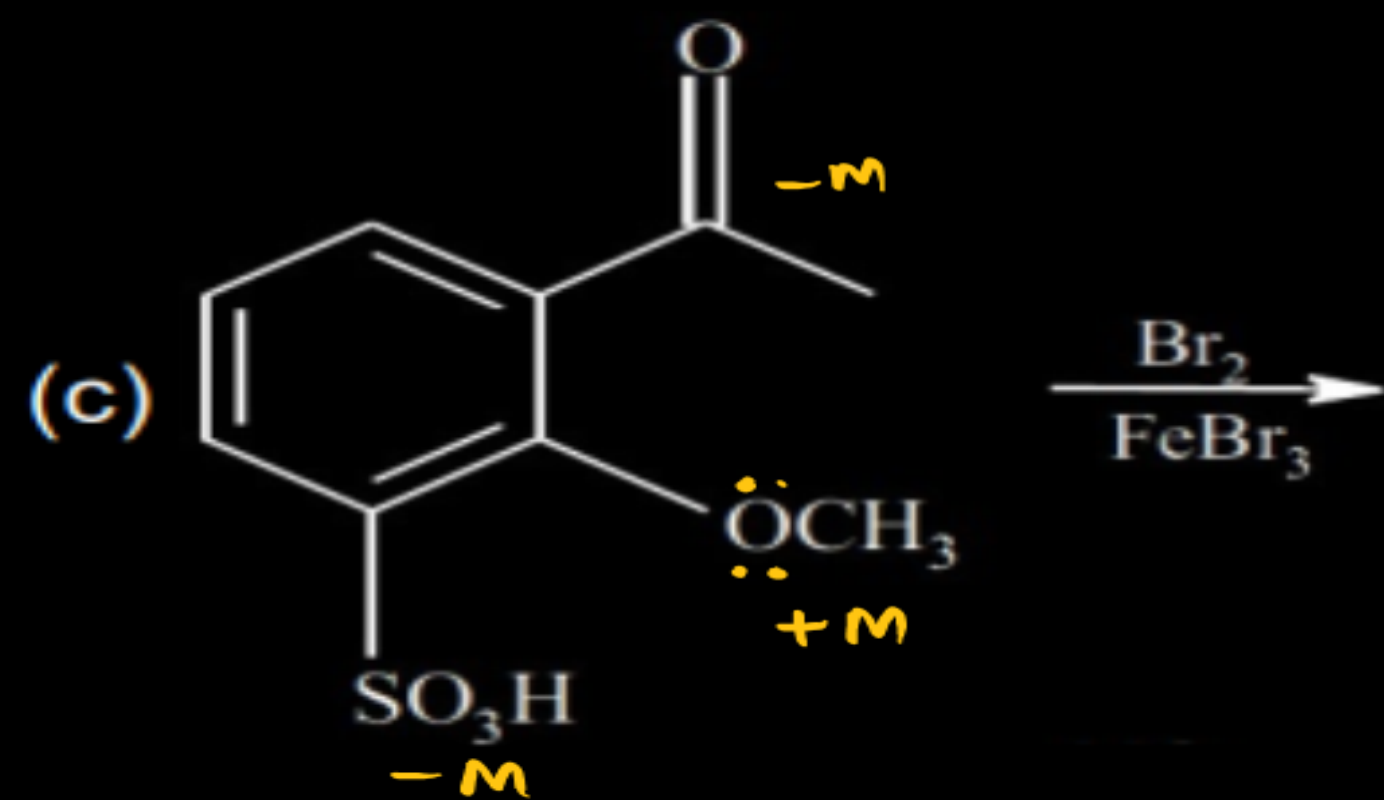
(A)



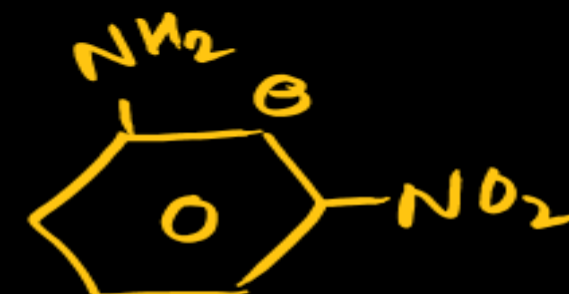


Predict major products in the following reactions:

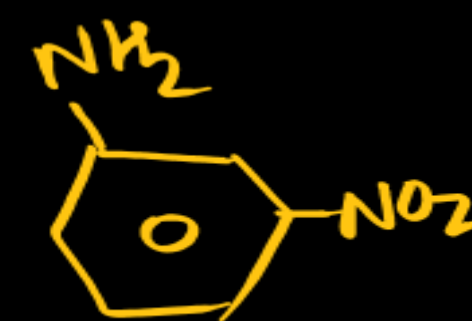


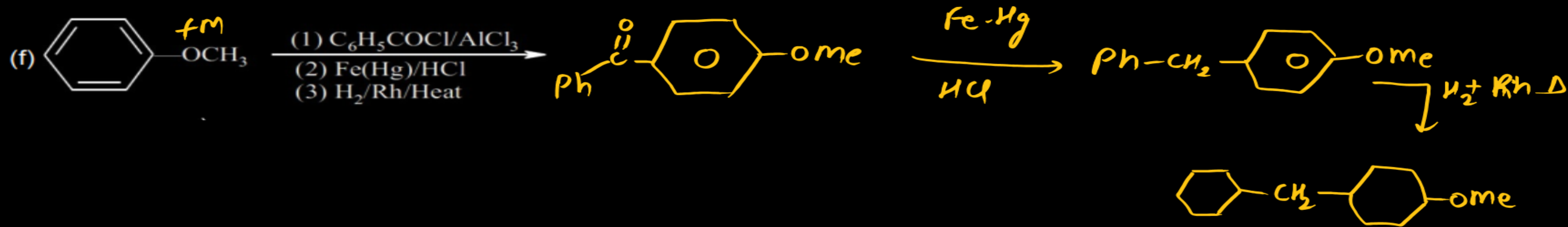
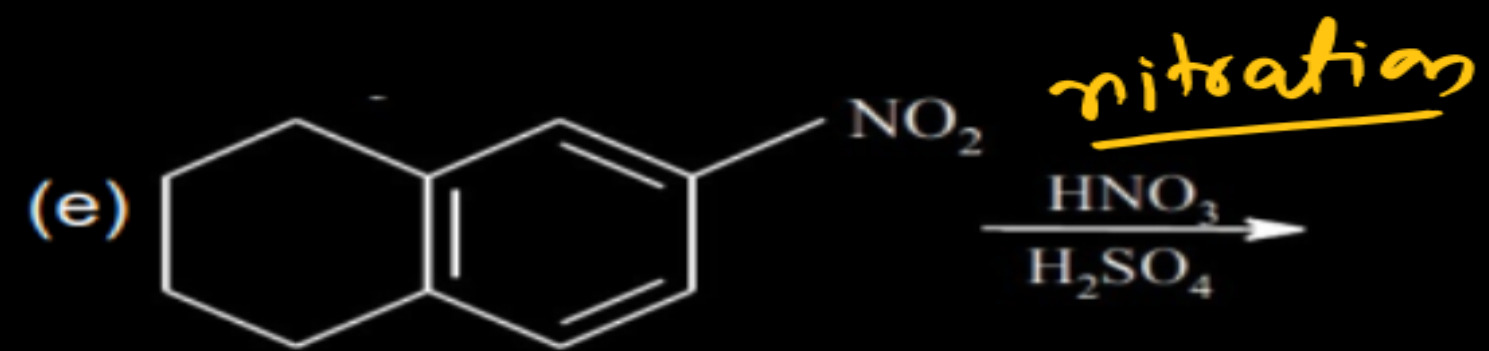


or



more stable





Identify the missing reagents/reactants/products in the following sequence of reactions:

